

02_data_exploration

May 31, 2026

```
[2]: import pandas as pd
import numpy as np

df = pd.read_csv('data/clean/db1b_agg.csv')
print(f"Loaded: {df.shape[0]:,} rows")
df.head()
```

Loaded: 176,405 rows

```
[2]:
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	Year	Quarter	Origin	Dest	OriginCityMarketID	DestCityMarketID	TkCarrier	\
0	2022	1	ABE	ATL	30135	30397	DL	
1	2022	1	ABE	BNA	30135	30693	G4	
2	2022	1	ABE	CLT	30135	31057	AA	
3	2022	1	ABE	DTW	30135	31295	DL	
4	2022	1	ABE	FLL	30135	32467	G4	

	Passengers	AvgFare	MedianFare	NonStopMiles	NumObs
0	446.0	208.256162	189.03	692.0	284
1	84.0	51.078431	47.00	685.0	51
2	206.0	233.775556	205.00	481.0	153
3	175.0	193.687910	160.00	425.0	134
4	329.0	81.002066	69.00	1041.0	121

```
[3]: import numpy as np
import pandas as pd

def cluster_robust_vcov(X, resid, cluster_ids, n_absorbed=0):
    """
    Cluster-robust (sandwich) covariance for OLS or within-FE OLS.
    X          : (N,k) regressors actually used in the final OLS. For FE models
                  these are the WITHIN-transformed regressors (no FE dummies).
    resid      : (N,) residuals from that OLS.
    cluster_ids: (N,) cluster label per row (here: route).
    n_absorbed: number of FE parameters partialled out (for the dof correction).
    """
    X = np.asarray(X, float); resid = np.asarray(resid, float)
    N, k = X.shape
    bread = np.linalg.inv(X.T @ X)
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clusters = np.unique(cluster_ids)
meat = np.zeros((k, k))
for g in clusters:
    m = cluster_ids == g
    s = X[m].T @ resid[m]          # (k,)
    meat += np.outer(s, s)
G = len(clusters)
k_total = k + n_absorbed
c = (G / (G - 1)) * ((N - 1) / (N - k_total))    # Stata-style small-sample
↪adj.
return c * (bread @ meat @ bread)

def demean(frame, value_cols, fe_cols, n_iter=100, tol=1e-12):
    """Iteratively demean value_cols by each FE dimension (within transform)."""
    work = frame[value_cols + fe_cols].copy().reset_index(drop=True)
    for _ in range(n_iter):
        maxch = 0.0
        for fe in fe_cols:
            for c in value_cols:
                gm = work.groupby(fe)[c].transform('mean')
                new = work[c] - gm
                maxch = max(maxch, float((new - work[c]).abs().max()))
                work[c] = new
        if maxch < tol:
            break
    return work[value_cols]

def run_fe(frame, value_cols, fe_cols, cluster_col='RouteID'):
    """OLS on within-demeaned data with cluster-robust SEs.
       value_cols[0] = dependent var; value_cols[1] = the DiD interaction."""
    dm = demean(frame, value_cols, fe_cols)
    y = dm[value_cols[0]].values
    X = dm[value_cols[1:]].values
    beta = np.linalg.lstsq(X, y, rcond=None)[0]
    resid = y - X @ beta
    n_abs = sum(frame[fe].nunique() for fe in fe_cols)
    V = cluster_robust_vcov(X, resid, frame[cluster_col].values,
↪n_absorbed=n_abs)
    se = np.sqrt(np.diag(V)); t = beta / se
    return dict(names=value_cols[1:], beta=beta, se=se, t=t, n=len(y),
                n_routes=frame['RouteID'].nunique())

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[4]: NK = 'NK'

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# Routes Spirit served in 2022 vs 2025
r2022 = df[(df.TkCarrier==NK)&(df.Year==2022)][['Origin', 'Dest']].
↪drop_duplicates()

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r2025 = df[(df.TkCarrier==NK)&(df.Year==2025)][['Origin', 'Dest']].
↳drop_duplicates()

# 176 exited routes (in 2022, not in 2025)
exited = r2022.merge(r2025, on=['Origin', 'Dest'], how='left', indicator=True)
exited = exited[exited._merge=='left_only'][['Origin', 'Dest']]
exited_set = set(map(tuple, exited.values))

# 86 "strong-exit" routes: Spirit avg >=500 pax/qtr in 2022, and exited
strong = (df[(df.TkCarrier==NK)&(df.Year==2022)]
          .groupby(['Origin', 'Dest']).Passengers.mean().reset_index())
strong = strong[strong.Passengers>=500][['Origin', 'Dest']]
strong = strong.merge(r2025, on=['Origin', 'Dest'], how='left', indicator=True)
strong = strong[strong._merge=='left_only'][['Origin', 'Dest']]
strong_set = set(map(tuple, strong.values))

ever_set = set(map(tuple, r2022.values))    # any Spirit presence in 2022

# Collapse non-Spirit carriers to the route-quarter level (passenger-weighted
↳fare)
o = df[df.TkCarrier!=NK].copy()
o['FareXPax'] = o.AvgFare * o.Passengers
g = (o.groupby(['Year', 'Quarter', 'Origin', 'Dest'])
     .agg(FareXPax=('FareXPax', 'sum'), TotalPax=('Passengers', 'sum'),
          NumCarriers=('TkCarrier', 'nunique'),
          ↳Distance=('NonStopMiles', 'first'))
     .reset_index())
g['WtdFare'] = g.FareXPax / g.TotalPax
g['LogFare'] = np.log(g.WtdFare)
g['LogPax'] = np.log(g.TotalPax + 1)
g['LogDist'] = np.log(g.Distance)
g['RouteID'] = g.Origin + '_' + g.Dest
g['rt'] = list(map(tuple, g[['Origin', 'Dest']].values))
g['SpiritExited'] = g.rt.isin(exited_set).astype(int)    # 176-route treatment
g['StrongExit'] = g.rt.isin(strong_set).astype(int)    # 86-route treatment
g['SpiritEver'] = g.rt.isin(ever_set).astype(int)
g['Control'] = (g.SpiritEver == 0).astype(int)
g['Post'] = ((g.Year>=2025)|((g.Year==2024)&(g.Quarter==4))).astype(int)
g['YQ_num'] = (g.Year-2022)*4 + g.Quarter
g['YQ'] = g.Year.astype(str)+'Q'+g.Quarter.astype(str)
g['Exit176_x_Post'] = g.SpiritExited * g.Post
g['Strong_x_Post'] = g.StrongExit * g.Post

# Matched control: never-Spirit routes with >=2 carriers and distance 400-2000
↳mi
rs = (g.groupby('RouteID').agg(avg_carr=('NumCarriers', 'mean'),
                              dist=('Distance', 'first')).reset_index())

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matched_routes = set(rs[(rs.avg_carr>=2)&(rs.dist>=400)&(rs.dist<=2000)].
↳RouteID)
g['MatchedControl'] = ((g.Control==1)&(g.RouteID.isin(matched_routes))).
↳astype(int)

df_reg = g
print(f"Route-quarter observations: {len(df_reg):,}")
print(f"Routes: {df_reg.RouteID.nunique():,} | Quarters: {df_reg.YQ.nunique():,}")
print(f"Exited (176) flagged route-qtrs: {df_reg.SpiritExited.sum():,}")
print(f"Strong-exit (86) flagged route-qtrs: {df_reg.StrongExit.sum():,}")
print(f"Strong-exit routes: {len(strong_set)} | Exited routes: {
↳len(exited_set)}")
print(f"Matched-control route-qtrs: {df_reg.MatchedControl.sum():,}")

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Route-quarter observations: 112,794
Routes: 12,795 | Quarters: 14
Exited (176) flagged route-qtrs: 1,915
Strong-exit (86) flagged route-qtrs: 1,092
Strong-exit routes: 86 | Exited routes: 176
Matched-control route-qtrs: 14,856

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[5]: m1 = df_reg
y = m1.LogFare.values
X = np.column_stack([np.ones(len(m1)), m1.SpiritExited, m1.Post,
↳m1.Exit176_x_Post, m1.LogDist, m1.LogPax])
names = ['Intercept', 'SpiritExited', 'Post', 'SpiritExited ×
↳Post', 'LogDist', 'LogPax']
beta = np.linalg.lstsq(X, y, rcond=None)[0]
resid = y - X @ beta
V = cluster_robust_vcov(X, resid, m1.RouteID.values, n_absorbed=0)
se = np.sqrt(np.diag(V)); t = beta / se

print("MODEL 1: OLS, 176-route treatment, cluster-robust SE (by route)")
print(f"{'Variable':<22}{'Coef':>10}{'ClustSE':>10}{'t':>8}")
for i,nm in enumerate(names):
    print(f"{'nm':<22}{'beta[i]:>10.4f'}{'se[i]:>10.4f'}{'t[i]:>8.2f}'")
b, s = beta[3], se[3]
print(f"\nDiD coef {b:+.4f} -> fare effect {(np.exp(b)-1)*100:+.1f}% "
↳f"| 95% CI [{(np.exp(b-1.96*s)-1)*100:+.1f}%, {(np.exp(b+1.96*s)-1)*100:+.
↳1f}%]")
m1_res = (b, s)

```

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MODEL 1: OLS, 176-route treatment, cluster-robust SE (by route)
Variable          Coef   ClustSE    t
Intercept         3.8665   0.0435   88.82
SpiritExited      -0.0509   0.0214   -2.38
Post              0.0283   0.0028   10.06
SpiritExited × Post -0.0152   0.0136   -1.11

```

LogDist	0.2190	0.0060	36.48
LogPax	0.0037	0.0012	3.16

DiD coef -0.0152 -> fare effect -1.5% | 95% CI [-4.1%, +1.2%]

```
[6]: # M2: route FE only, 86-route treatment (Post stays in; not absorbed)
M2 = run_fe(df_reg, ['LogFare', 'Strong_x_Post', 'Post', 'LogPax'], ['RouteID'])

# M3 (= paper's "Model 2"): route + quarter FE, 86-route treatment
M3 = run_fe(df_reg, ['LogFare', 'Strong_x_Post', 'LogPax'], ['RouteID', 'YQ'])

# M4: route + quarter FE, 86-route treatment, MATCHED control only
m4_sample = df_reg[(df_reg.StrongExit==1) | (df_reg.MatchedControl==1)].copy()
M4 = run_fe(m4_sample, ['LogFare', 'Strong_x_Post', 'LogPax'], ['RouteID', 'YQ'])

def line(tag, res):
    b, s = res['beta'][0], res['se'][0]
    pct = (np.exp(b)-1)*100
    lo = (np.exp(b-1.96*s)-1)*100
    hi = (np.exp(b+1.96*s)-1)*100
    print(f"{tag:<34}{b:>9.4f}{s:>9.4f}{res['t'][0]:>7.2f}{pct:>8.1f}%"
          f"    [{lo:+.1f}%, {hi:+.1f}%]    N={res['n']:,}")

b1,s1 = m1_res
pct1=(np.exp(b1)-1)*100; lo1=(np.exp(b1-1.96*s1)-1)*100; hi1=(np.exp(b1+1.
↪96*s1)-1)*100
print("Effect of Spirit exit on competitor fares (cluster-robust SE by route)")
print(f"{'Model':<34}{'Coef':>9}{'SE':>9}{'t':>7}{'%Δ':>8}    95% CI")
print("-"*88)
print(f"{'M1: OLS, 176 routes, no FE':<34}{b1:>9.4f}{s1:>9.4f}{b1/s1:>7.2f}"
      f"{'pct1:>8.1f}%    [{lo1:+.1f}%, {hi1:+.1f}%]")
line('M2: Route FE, 86 routes', M2)
line('M3: Route+Qtr FE, 86 routes', M3)
line('M4: Route+Qtr FE, 86, matched ctrl', M4)
```

Effect of Spirit exit on competitor fares (cluster-robust SE by route)

Model	Coef	SE	t	%Δ	95% CI
M1: OLS, 176 routes, no FE	-0.0152	0.0136	-1.11	-1.5%	[-4.1%, +1.2%]
M2: Route FE, 86 routes	0.0131	0.0117	1.12	1.3%	[-1.0%, +3.7%]
M3: Route+Qtr FE, 86 routes	0.0129	0.0117	1.11	1.3%	[-1.0%, +3.6%]
M4: Route+Qtr FE, 86, matched ctrl	0.0063	0.0120	0.53	0.6%	[-1.7%, +3.0%]

```

[7]: import matplotlib.pyplot as plt

REF = 11 # YQ_num for 2024Q3, the quarter just BEFORE the 2024Q4 filing
es = df_reg.copy()
es_cols, es_map = [], {}
for q in sorted(es.YQ_num.unique()):
    if q == REF:
        continue
    col = f'ES_{q}'
    es[col] = es.StrongExit * (es.YQ_num == q).astype(int)
    es_cols.append(col); es_map[col] = q

ES = run_fe(es, ['LogFare'] + es_cols + ['LogPax'], ['RouteID', 'YQ'])
coef = dict(zip(ES['names'], ES['beta']))
sek = dict(zip(ES['names'], ES['se']))

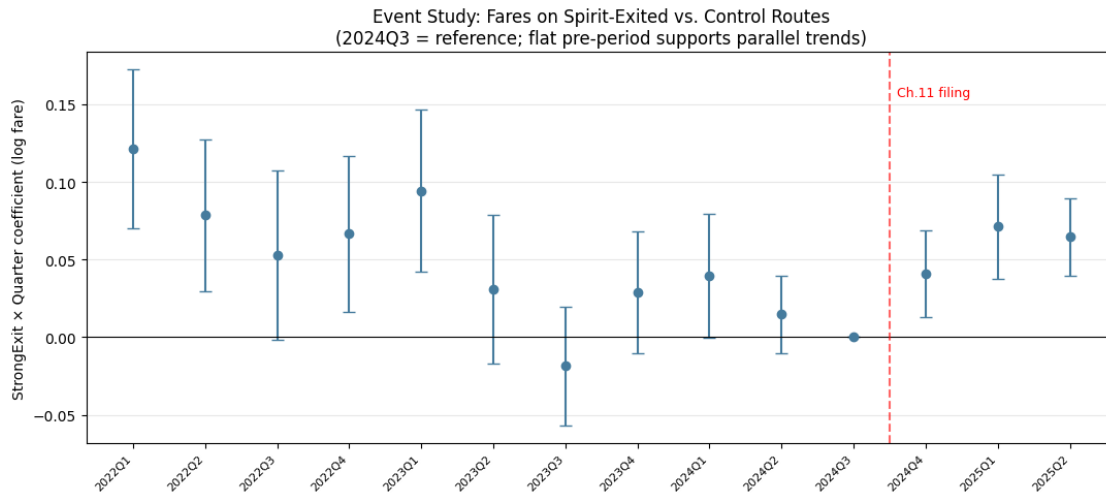
# assemble for plotting (reference quarter pinned at 0)
xs = sorted(es.YQ_num.unique())
pts, los, his = [], [], []
for q in xs:
    if q == REF:
        pts.append(0); los.append(0); his.append(0)
    else:
        col = f'ES_{q}'
        b, s = coef[col], sek[col]
        pts.append(b); los.append(b-1.96*s); his.append(b+1.96*s)

labels = (es[['YQ_num', 'YQ']].drop_duplicates().sort_values('YQ_num').YQ.
    ↪tolist())
fig, ax = plt.subplots(figsize=(11,5))
ax.errorbar(xs, pts, yerr=[np.array(pts)-np.array(los), np.array(his)-np.
    ↪array(pts)],
            fmt='o', color='#457b9d', capsize=4, linewidth=1.5)
ax.axhline(0, color='black', lw=0.8)
ax.axvline(11.5, color='red', ls='--', alpha=0.6)
ax.text(11.6, max(his)*0.9, 'Ch.11 filing', color='red', fontsize=9)
ax.set_xticks(xs); ax.set_xticklabels(labels, rotation=45, ha='right',
    ↪fontsize=8)
ax.set_ylabel('StrongExit × Quarter coefficient (log fare)')
ax.set_title('Event Study: Fares on Spirit-Exited vs. Control Routes\n'
            '(2024Q3 = reference; flat pre-period supports parallel trends)')
ax.grid(axis='y', alpha=0.3)
plt.tight_layout(); plt.savefig('figure_event_study.png', dpi=150); plt.show()

print("Pre-period (should be ~0 and individually insignificant if PT holds):")
for q in xs:
    if q < REF and q != REF:

```

```
col=f'ES_{q}'; b,s=coef[col],sek[col]
flag = '' if abs(b/s)<1.96 else ' <-- significant pre-trend!'
print(f" {labels[xs.index(q)]}: {b:+.4f} (t={b/s:+.2f}){flag}")
```



Pre-period (should be ~0 and individually insignificant if PT holds):

```
2022Q1: +0.1211 (t=+4.63) <-- significant pre-trend!
2022Q2: +0.0785 (t=+3.15) <-- significant pre-trend!
2022Q3: +0.0528 (t=+1.90)
2022Q4: +0.0665 (t=+2.59) <-- significant pre-trend!
2023Q1: +0.0942 (t=+3.54) <-- significant pre-trend!
2023Q2: +0.0306 (t=+1.25)
2023Q3: -0.0186 (t=-0.96)
2023Q4: +0.0289 (t=+1.44)
2024Q1: +0.0394 (t=+1.94)
2024Q2: +0.0148 (t=+1.17)
```

```
[8]: d = df[df.TkCarrier!='NK'].copy()
d['rt'] = list(map(tuple, d[['Origin','Dest']].values))
d['Period'] = np.where((d.Year>=2025)|((d.Year==2024)&(d.
    ↳Quarter==4)), 'Post', 'Pre')
d['grp'] = np.where(d.rt.isin(exited_set), 'Exited',
    np.where(d.rt.isin(ever_set), 'Stayed', 'Control'))

def pre_post(carrier, grp):
    sub = d[(d.TkCarrier==carrier)&(d.grp==grp)]
    pre = sub[sub.Period=='Pre'].Passengers.mean()
    post = sub[sub.Period=='Post'].Passengers.mean()
    return pre, post

print("Carrier growth on EXITED routes vs. the same carrier on CONTROL routes")
```

```

print(f"{'Carr':<6}{'ExitedΔ%':>10}{'ControlΔ%':>11}{'DiD (pp)':>10}")
print("-"*40)
for c in ['DL', 'AA', 'WN', 'F9', 'B6']:
    ep, eo = pre_post(c, 'Exited'); cp, co = pre_post(c, 'Control')
    e_pct = (eo/ep-1)*100 if ep else np.nan
    c_pct = (co/cp-1)*100 if cp else np.nan
    print(f"{'c':<6}{'e_pct':>9.1f}%{'c_pct':>10.1f}%{'e_pct-c_pct':>9.1f}")

# Extensive-margin check you already had: non-Spirit carrier COUNT DiD
cc = (d.groupby(['Year', 'Quarter', 'rt', 'grp']).TkCarrier.nunique()
      .reset_index().rename(columns={'TkCarrier': 'NumCarr'}))
cc['Period'] = np.where((cc.Year>=2025)|((cc.Year==2024)&(cc.
    ↳Quarter==4)), 'Post', 'Pre')
tab = cc[cc.grp.isin(['Exited', 'Control'])].groupby(['grp', 'Period']).NumCarr.
    ↳mean().unstack()
did = (tab.loc['Exited', 'Post']-tab.loc['Exited', 'Pre']) - \
      (tab.loc['Control', 'Post']-tab.loc['Control', 'Pre'])
print(f"\nNon-Spirit carrier-count DiD: {did:+.3f} "
      f"(near 0 => little NEW entry; backfill was intensive-margin)")
print(tab.round(2).to_string())

```

Carrier growth on EXITED routes vs. the same carrier on CONTROL routes

Carr	ExitedΔ%	ControlΔ%	DiD (pp)
------	----------	-----------	----------

DL	15.3%	3.1%	12.3
AA	16.3%	9.1%	7.2
WN	13.2%	31.2%	-17.9
F9	6.7%	-8.6%	15.2
B6	4.9%	38.7%	-33.8

Non-Spirit carrier-count DiD: -0.002 (near 0 => little NEW entry; backfill was intensive-margin)

Period	Post	Pre
grp		
Control	1.45	1.43
Exited	2.20	2.18

```

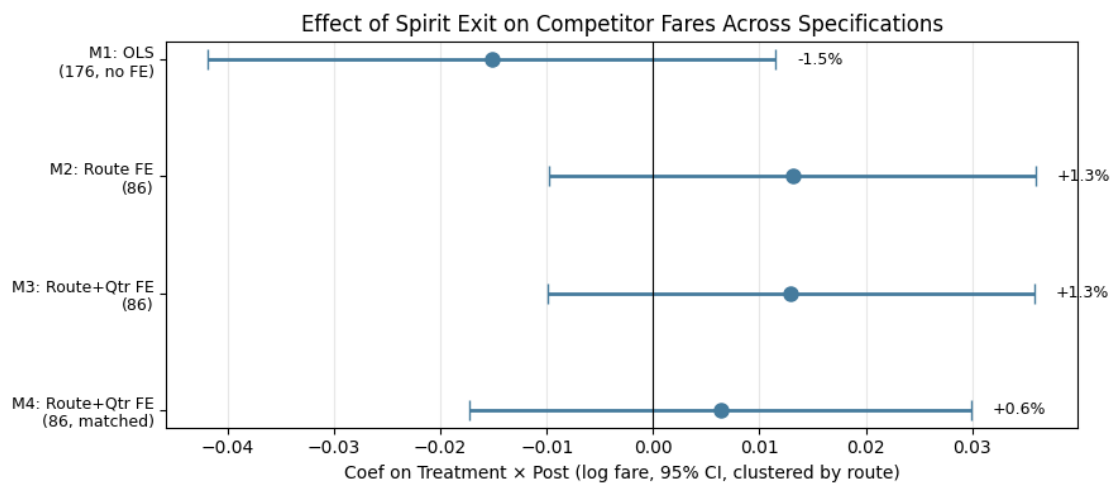
[9]: results = {
    'M1: OLS\n(176, no FE)': m1_res,
    'M2: Route FE\n(86)': (M2['beta'][0], M2['se'][0]),
    'M3: Route+Qtr FE\n(86)': (M3['beta'][0], M3['se'][0]),
    'M4: Route+Qtr FE\n(86, matched)': (M4['beta'][0], M4['se'][0]),
}
labels = list(results.keys())
coefs = [v[0] for v in results.values()]
ses = [v[1] for v in results.values()]

```

```

fig, ax = plt.subplots(figsize=(9,4))
yp = range(len(labels))
ax.errorbar(coefs, yp, xerr=[1.96*s for s in ses], fmt='o', color='#457b9d',
            markersize=8, capsize=6, linewidth=2)
ax.axvline(0, color='black', lw=0.8)
ax.set_yticks(list(yp)); ax.set_yticklabels(labels, fontsize=9)
ax.set_xlabel('Coef on Treatment × Post (log fare, 95% CI, clustered by route)')
ax.set_title('Effect of Spirit Exit on Competitor Fares Across Specifications')
for i,(c,s) in enumerate(zip(coefs,ses)):
    ax.text(c+1.96*s+0.002, i, f'{(np.exp(c)-1)*100:+.1f}%', va='center',
            ↪ fontsize=9)
ax.invert_yaxis(); ax.grid(axis='x', alpha=0.3)
plt.tight_layout(); plt.savefig('figure3_coefficients.png', dpi=150); plt.show()

```



```

[10]: # Figure 1: Fare Trends (Indexed to 2022 Q1)
# Passenger-weighted average fare by group and quarter for non-Spirit carriers

df_viz = df[df.TkCarrier != 'NK'].copy()
df_viz['rt'] = list(map(tuple, df_viz[['Origin', 'Dest']].values))
df_viz['Group'] = np.where(df_viz.rt.isin(exited_set), 'Spirit exited',
                           np.where(df_viz.rt.isin(ever_set), 'Spirit stayed',
                                     'Never served by Spirit'))
df_viz['YQ_num'] = (df_viz.Year - 2022) * 4 + df_viz.Quarter
df_viz['YQ'] = df_viz.Year.astype(str) + 'Q' + df_viz.Quarter.astype(str)

trends = (df_viz.groupby(['YQ_num', 'YQ', 'Group'])
          .apply(lambda g: pd.Series({
              'WtdAvgFare': (g.AvgFare * g.Passengers).sum() / g.Passengers.
              ↪ sum()
          })).reset_index())

```

```

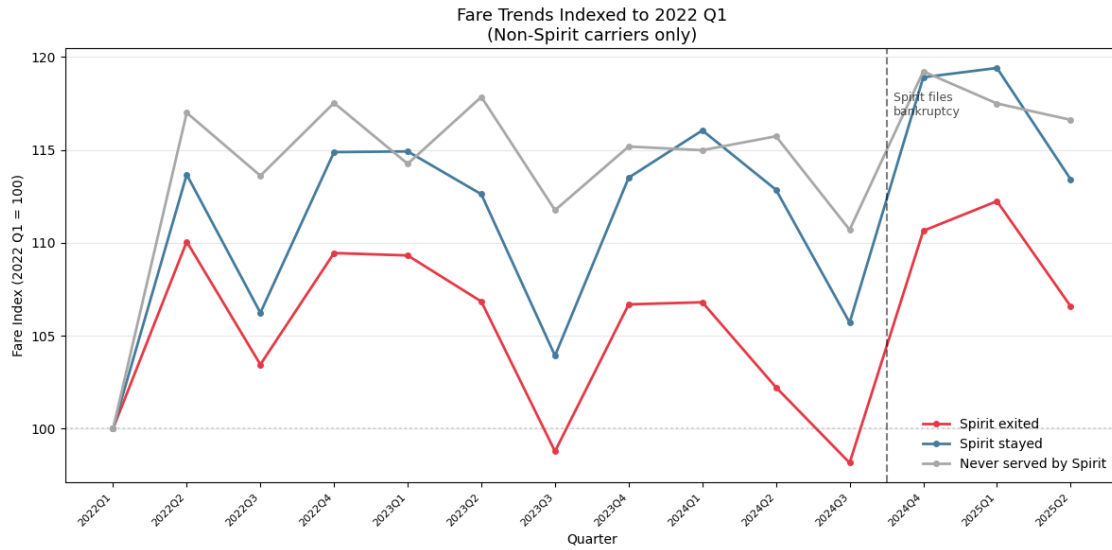
colors = {'Spirit exited': '#e63946',
          'Spirit stayed': '#457b9d',
          'Never served by Spirit': '#a8a8a8'}

fig, ax = plt.subplots(figsize=(12, 6))
for group in ['Spirit exited', 'Spirit stayed', 'Never served by Spirit']:
    gdata = trends[trends.Group == group].sort_values('YQ_num').copy()
    base = gdata.WtdAvgFare.iloc[0]
    gdata['Indexed'] = (gdata.WtdAvgFare / base) * 100
    ax.plot(gdata.YQ_num, gdata.Indexed, marker='o', markersize=4,
            linewidth=2, color=colors[group], label=group)

ax.axvline(x=11.5, color='black', linestyle='--', alpha=0.5)
ax.text(11.6, ax.get_ylim()[1] * 0.97, 'Spirit files\nbankruptcy', fontsize=9,
        alpha=0.7)
ax.axhline(y=100, color='gray', linestyle=':', alpha=0.3)
ax.set_xlabel('Quarter')
ax.set_ylabel('Fare Index (2022 Q1 = 100)')
ax.set_title('Fare Trends Indexed to 2022 Q1\n(Non-Spirit carriers only)',
             alpha=0.7)
xticks = trends[['YQ_num', 'YQ']].drop_duplicates().sort_values('YQ_num')
ax.set_xticks(xticks.YQ_num); ax.set_xticklabels(xticks.YQ, rotation=45,
        ha='right', fontsize=8)
ax.legend(frameon=False); ax.grid(axis='y', alpha=0.3)
plt.tight_layout(); plt.savefig('figure1_indexed_trends.png', dpi=150); plt.
    show()

# Print exact indexed values for reference
for group in ['Spirit exited', 'Spirit stayed', 'Never served by Spirit']:
    gdata = trends[trends.Group == group].sort_values('YQ_num').copy()
    base = gdata.WtdAvgFare.iloc[0]
    gdata['Indexed'] = (gdata.WtdAvgFare / base) * 100
    print(f"\n {group} ")
    for _, row in gdata.iterrows():
        print(f" {row.YQ}: {row.Indexed:.1f}")

```



Spirit exited

2022Q1: 100.0
 2022Q2: 110.1
 2022Q3: 103.4
 2022Q4: 109.4
 2023Q1: 109.3
 2023Q2: 106.8
 2023Q3: 98.8
 2023Q4: 106.7
 2024Q1: 106.8
 2024Q2: 102.2
 2024Q3: 98.2
 2024Q4: 110.6
 2025Q1: 112.2
 2025Q2: 106.6

Spirit stayed

2022Q1: 100.0
 2022Q2: 113.7
 2022Q3: 106.2
 2022Q4: 114.9
 2023Q1: 114.9
 2023Q2: 112.6
 2023Q3: 103.9
 2023Q4: 113.5
 2024Q1: 116.0
 2024Q2: 112.9
 2024Q3: 105.7

2024Q4: 118.9
2025Q1: 119.4
2025Q2: 113.4

Never served by Spirit

2022Q1: 100.0
2022Q2: 117.0
2022Q3: 113.6
2022Q4: 117.5
2023Q1: 114.3
2023Q2: 117.9
2023Q3: 111.8
2023Q4: 115.2
2024Q1: 115.0
2024Q2: 115.7
2024Q3: 110.7
2024Q4: 119.2
2025Q1: 117.5
2025Q2: 116.6

[11]: # *Figure 3: Spirit Airlines' Domestic Operations*

```
spirit_data = df[df.TkCarrier == 'NK'].copy()
spirit_data['YQ_num'] = (spirit_data.Year - 2022) * 4 + spirit_data.Quarter
spirit_data['YQ'] = spirit_data.Year.astype(str) + 'Q' + spirit_data.Quarter.
    ↳astype(str)

spirit_quarterly = []
for (yq_num, yq), group in spirit_data.groupby(['YQ_num', 'YQ']):
    n_routes = group[['Origin', 'Dest']].drop_duplicates().shape[0]
    total_pax = group.Passengers.sum()
    spirit_quarterly.append({'YQ_num': yq_num, 'YQ': yq,
                           'NumRoutes': n_routes, 'TotalPax': total_pax})
spirit_qtr = pd.DataFrame(spirit_quarterly).sort_values('YQ_num')

fig, ax1 = plt.subplots(figsize=(12, 5))
color1, color2 = '#e63946', '#457b9d'

ax1.bar(spirit_qtr.YQ_num, spirit_qtr.NumRoutes, color=color1, alpha=0.7,
    ↳label='Routes served')
ax1.set_ylabel('Number of Nonstop Routes', color=color1, fontsize=11)
ax1.tick_params(axis='y', labelcolor=color1)

ax2 = ax1.twinx()
ax2.plot(spirit_qtr.YQ_num, spirit_qtr.TotalPax / 1000,
        color=color2, marker='o', linewidth=2.5, label='Passengers (000s)')
ax2.set_ylabel('Passengers (thousands)', color=color2, fontsize=11)
```

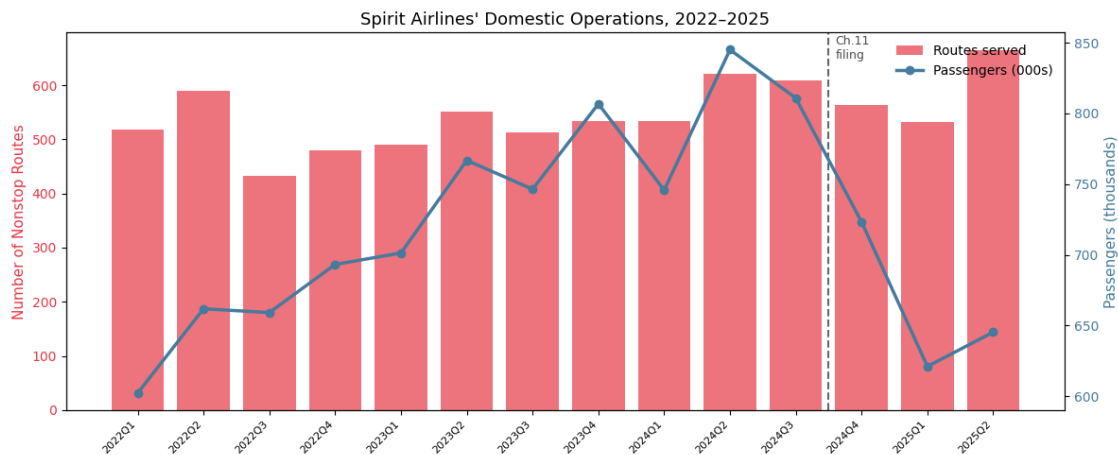
```

ax2.tick_params(axis='y', labelcolor=color2)

ax1.axvline(x=11.5, color='black', linestyle='--', alpha=0.6)
ax1.text(11.6, ax1.get_ylim()[1] * 0.93, 'Ch.11\nfiling', fontsize=9, alpha=0.7)
ax1.set_xticks(spirit_qtr.YQ_num.tolist())
ax1.set_xticklabels(spirit_qtr.YQ.tolist(), rotation=45, ha='right', fontsize=8)
ax1.set_title("Spirit Airlines' Domestic Operations, 2022-2025", fontsize=13)
lines1, labels1 = ax1.get_legend_handles_labels()
lines2, labels2 = ax2.get_legend_handles_labels()
ax1.legend(lines1 + lines2, labels1 + labels2, loc='upper right', frameon=False)
plt.tight_layout(); plt.savefig('figure3_spirit_footprint.png', dpi=150); plt.
    show()

# Print quarterly summary table
print("\n Spirit Airlines by Quarter \n")
print(f"{'Quarter':<10} {'Routes':>8} {'Passengers':>12}")
print("-" * 32)
for _, row in spirit_qtr.iterrows():
    print(f"{'row.YQ':<10} {'row.NumRoutes':>8} {'row.TotalPax':>12,.0f}")

```



Spirit Airlines by Quarter

Quarter	Routes	Passengers

2022Q1	518	602,326
2022Q2	589	661,857
2022Q3	432	659,192
2022Q4	480	693,220
2023Q1	491	701,335
2023Q2	551	766,851

2023Q3	513	746,463
2023Q4	534	806,807
2024Q1	533	745,795
2024Q2	621	845,104
2024Q3	608	810,902
2024Q4	563	723,251
2025Q1	532	621,093
2025Q2	664	645,335

```
[12]: # Table 3: Carrier Backfill Detail
# Which specific carriers grew on Spirit-exited routes? (absolute numbers)

d_backfill = df[df.TkCarrier != 'NK'].copy()
d_backfill['rt'] = list(map(tuple, d_backfill[['Origin', 'Dest']].values))
d_backfill['Period'] = np.where(
    (d_backfill.Year >= 2025) | ((d_backfill.Year == 2024) & (d_backfill.
    ↪Quarter == 4)),
    'Post', 'Pre')

df_exited_only = d_backfill[d_backfill.rt.isin(exited_set)]

carrier_growth = (
    df_exited_only
    .groupby(['TkCarrier', 'Period'])
    .agg(AvgQtrPax=('Passengers', 'mean'))
    .reset_index()
    .pivot(index='TkCarrier', columns='Period', values='AvgQtrPax')
    .fillna(0)
)

carrier_growth['Change'] = carrier_growth['Post'] - carrier_growth['Pre']
carrier_growth['PctChange'] = ((carrier_growth['Post'] / carrier_growth['Pre']) -
    ↪1) * 100
carrier_growth = carrier_growth.sort_values('Change', ascending=False)

print(" Which Carriers Grew on Spirit-Exited Routes? ")
print("(Average quarterly passengers per route, pre vs post bankruptcy)\n")
print(f"{'Carrier':<10} {'Pre':>10} {'Post':>10} {'Change':>10} {'% Change':
    ↪>10}")
print("-" * 55)
for carrier in carrier_growth.head(10).index:
    row = carrier_growth.loc[carrier]
    print(f"{'carrier':<10} {row['Pre']:>10.0f} {row['Post']:>10.0f} "
          f"{row['Change']:>10.0f} {row['PctChange']:>9.1f}%")
print(f"\n{'TOTAL':<10} {carrier_growth['Pre'].sum():>10.0f} "
      f"{carrier_growth['Post'].sum():>10.0f} {carrier_growth['Change'].sum():
    ↪>10.0f}")
```

Which Carriers Grew on Spirit-Exited Routes?

(Average quarterly passengers per route, pre vs post bankruptcy)

Carrier	Pre	Post	Change	% Change
DL	2455	2832	+377	+15.3%
AA	1067	1240	+174	+16.3%
WN	1119	1268	+148	+13.2%
F9	1353	1443	+90	+6.7%
B6	1410	1479	+69	+4.9%
SY	1946	1992	+46	+2.3%
MX	330	371	+41	+12.5%
G4	117	122	+5	+4.6%
AC	1	2	+1	+100.0%
PR	1	1	+0	+0.0%
TOTAL	12283	12937	+654	